

Fixed Point Theorems in Banach Algebra

and

Lattice-Theoretic Fixed Point Theorems

Iterative Methods and Approximation (Sections 5.8–5.9)

Based on Pathak: An Introduction to Nonlinear Analysis and Fixed Point Theory

July 14, 2026

Overview

- 1 Contraction Mappings and Lipschitzian Operators
- 2 Fixed Point Theorems in Banach Algebra
- 3 Lattice-Theoretic Fixed Point Theorems
- 4 Integration with Banach Algebra Theory
- 5 Applications and Computational Methods
- 6 Summary and Extensions

Fundamental Concepts

Definition (Nonlinear Contraction - Definition 5.92)

A mapping $T : X \rightarrow X$ on a Banach space X is a **nonlinear contraction** if:

$$\|Tx - Ty\| \leq \phi(\|x - y\|), \quad \forall x, y \in X,$$

where $\phi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is continuous with $\phi(r) < r$ for all $r > 0$.

Definition ($\mathcal{D}$ -Lipschitzian Mappings - Definition 5.93)

A mapping $T : X \rightarrow X$ is **$\mathcal{D}$ -Lipschitzian** (Dhage, 2003) if:

$$\|Tx - Ty\| \leq \phi(\|x - y\|), \quad \forall x, y \in X,$$

where $\phi : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is continuous and nondecreasing with $\phi(0) = 0$.

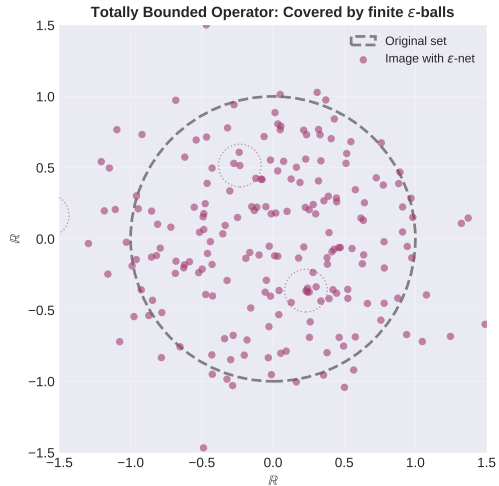
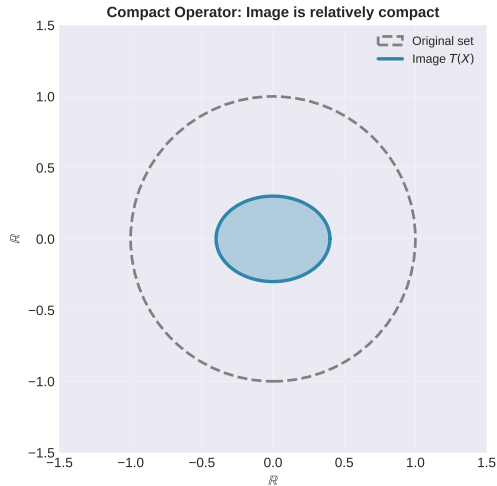
Key difference: Nonlinear contractions require $\phi(r) < r$, while $\mathcal{D}$ -Lipschitzian mappings only require $\phi(0) = 0$.

Classification of Operators

- **Compact Operator:** $T(X)$ is a compact subset of X
 - Every bounded sequence in $T(X)$ has a convergent subsequence
- **Totally Bounded Operator:** For any bounded $S \subseteq X$, $T(S)$ is totally bounded
 - Can be covered by finitely many ε -balls for any $\varepsilon > 0$
- **Completely Continuous Operator:** Both continuous and totally bounded
 - Every compact operator is completely continuous
 - But converse may not hold

Relationship: Compact $\Rightarrow$ Totally bounded $\Rightarrow$ Completely continuous

Illustration of Operator Types



Operator Classification Hierarchy

Krasnoselski Fixed Point Theorem I

Theorem (Krasnoselski - Theorem 5.176)

Let S be a closed, convex, and bounded subset of a Banach algebra X . Let $A, B : S \rightarrow X$ be two operators such that:

- (a) A is a contraction;
- (b) B is completely continuous; and
- (c) $Ax + By \in S, \forall x, y \in S$.

Then the operator equation

$$Ax + Bx = x$$

has a solution in S .

Significance:

- Decomposes the fixed point problem into a contraction and a compact perturbation
- Enables treatment of more complex operator equations
- Foundation for many applications in differential and integral equations

Burton's Modification

Theorem (Burton - Theorem 5.177)

Let S be a closed, convex, and bounded subset of a Banach algebra X . Let $A : X \rightarrow X$ and $B : S \rightarrow X$ be two operators such that:

- (a) A is a contraction;*
- (b) [Weaker condition] $B : S \rightarrow X$ satisfies a mild boundedness condition; and*
- (c) [Weaker covering] A milder covering condition replacing (c) from Krasnoselski.*

Then $Ax + Bx = x$ has a solution in S .

Key Innovation:

- Burton weakened Krasnoselski's stringent hypothesis (c)
- B need not be completely continuous—only bounded
- More applicable to practical problems

$\mathcal{P}$ -Lipschitzian Fixed Point Theorem

Theorem (Theorem 5.184)

Let S be a closed, convex, and bounded subset of a Banach space X . Let $A : X \rightarrow X$ and $B : S \rightarrow X$ be two operators such that:

- (a) A is a $\mathcal{P}$ -Lipschitzian contraction;
- (b) B is completely continuous; and
- (c) $\|x\| \leq \|Ax + By\|, \forall x \in X, y \in S$.

If $M\phi(r) + \phi_C(r) < r$ for $r > 0$, where $M = \|B(S)\|$, then $Ax + Bx = x$ has a solution, whenever $M\phi(r) < r$.

Listing: Verification of $\mathcal{P}$ -Lipschitzian condition

```
# Example: Checking contractiveness condition
import numpy as np

def phi(r):
    """Example  $\mathcal{P}$ -Lipschitzian function"""
    return 0.5 * r # Linear case

r_values = np.linspace(0.1, 10, 100)
M = 2.0 # Bound on operator B
```

$\mathcal{D}$ -Lipschitzian Fixed Point Theorem

Theorem (Theorem 5.185)

Let S be a closed, convex, and bounded subset of a Banach algebra X . Let $A : X \rightarrow X$ and $B : S \rightarrow X$ be two operators such that:

- (a) A is $\mathcal{D}$ -Lipschitzian;
- (b) B is completely continuous; and
- (c) $x = AxBy \Rightarrow x \in S, \forall y \in S$.

Then the operator equation $AxBy = x$ has a solution whenever $M\phi(r) < r, r > 0$, where $M = \|B(S)\|$.

Definition (Sufficient Condition - Proposition 5.14)

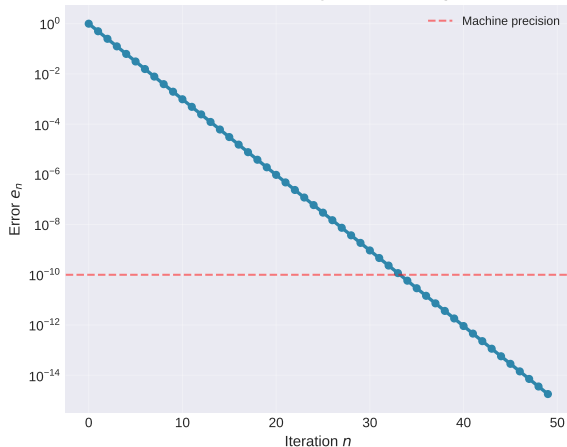
If $S = \{y \in X : \|y\| \leq r\}$ and

$$\|x\| \leq \|(I - C)x\|, \quad \forall x \in X,$$

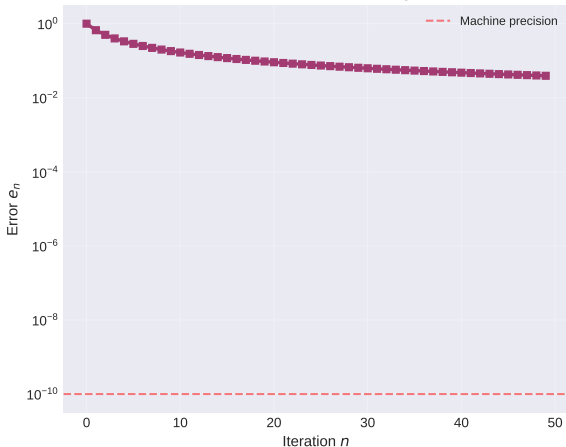
then hypothesis (c) is satisfied in Theorem 5.177.

Convergence and Iteration

Banach Contraction: Exponential Convergence



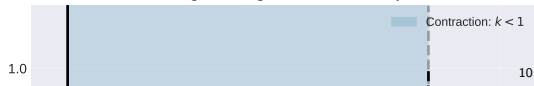
Mann Iteration: Slower Convergence



Convergence Rate Comparison



Convergence Region in Parameter Space



Numerical Example: Computing Fixed Points

Listing: Fixed point iteration for Krasnoselski operator

```
import numpy as np
from numpy.linalg import norm

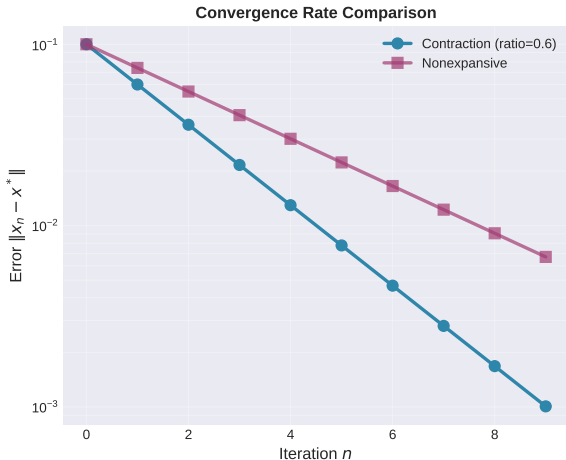
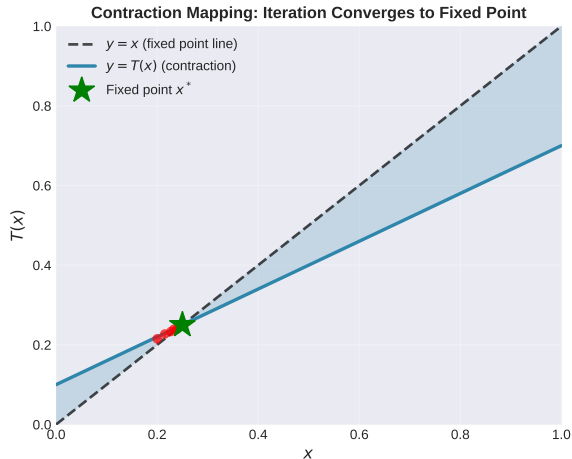
def contraction_A(x, k=0.5):
    """Contraction operator:  $A(x) = kx$ """
    return k * x

def compact_B(x, epsilon=0.1):
    """Compact perturbation:  $B(x) = \epsilon \sin(x)$ """
    return epsilon * np.sin(x)

def krasnoselski_iterate(x, alpha=0.5):
    """Krasnoselski iteration:  $x_{n+1} = A(x_n) + B(x_n)$ """
    return contraction_A(x) + compact_B(x)

# Fixed point iteration
x = np.array([1.0, 2.0, 3.0])
errors = []
for n in range(20):
    x_old = x.copy()
    x = krasnoselski_iterate(x)
    error = norm(x - x_old)
    errors.append(error)
    if n < 5 or n >= 18:
        print(f"Iter {n:2d}: ||x_new - x_old|| = {error:.6e}")
```

Relationship Between Operator Types



Key Insights:

- Contraction mapping: Every iteration brings points closer to fixed point
- Fixed point is unique and guaranteed to exist (Banach Contraction Principle)

Ordered Banach Spaces and Lattices

Definition (Partially Ordered Set)

A set X with a binary relation $\leq$ is **partially ordered** if for all $x, y, z \in X$:

- ① Reflexivity: $x \leq x$
- ② Antisymmetry: $x \leq y$ and $y \leq x \Rightarrow x = y$
- ③ Transitivity: $x \leq y$ and $y \leq z \Rightarrow x \leq z$

Definition (Complete Lattice)

A poset L is a **complete lattice** if every subset has both a least upper bound (supremum) and a greatest lower bound (infimum).

Examples:

- Power set 2^X (all subsets of X)
- Intervals in $\mathbb{R}$: $[a, b]$ with usual ordering
- Banach lattices: Normed vector lattices with order-continuous norm

Knaster-Tarski Fixed Point Theorem I

Theorem (Knaster-Tarski (1927))

Let $f : L \rightarrow L$ be a monotone increasing function on a complete lattice L . Then:

- 1 The set of fixed points $\text{Fix}(f) = \{x \in L : f(x) = x\}$ is non-empty.
- 2 The set of fixed points itself forms a complete lattice.
- 3 The function f has a least fixed point and a greatest fixed point.

Key Advantage:

- Does **not** require continuity of f
- Applies to discrete structures and infinite-dimensional spaces
- Guarantees existence without constructive iteration

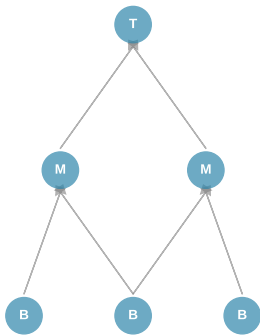
Knaster-Tarski Fixed Point Theorem II

Proof Strategy:

- Consider the set of prefixed points: $A = \{x \in L : x \leq f(x)\}$
- If $x \leq f(x)$, then $f(x) \leq f(f(x))$ (by monotonicity)
- The supremum $x^* = \sup A$ satisfies $f(x^*) = x^*$
- This x^* is the least fixed point
- Similarly, there exists a greatest fixed point

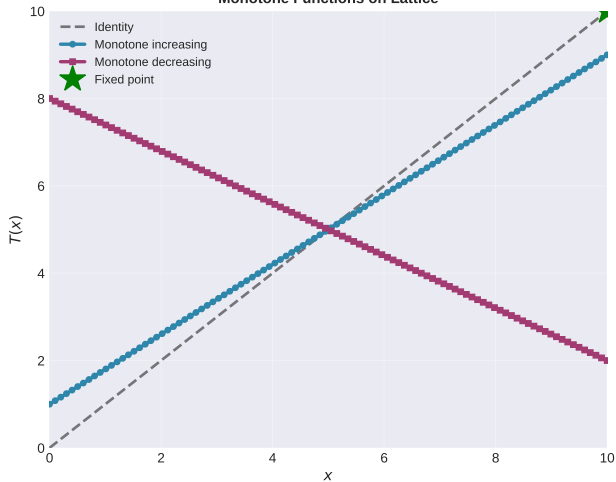
Monotone Functions and Fixed Points

Partially Ordered Set (Poset)



Knaster-Tarski Fixed Point Theorem

Monotone Functions on Lattice



Reflexivity in Banach Lattices

Applications of Lattice Fixed Points

Computer Science Applications:

- **Program Verification:** Computing fixed points of program transformations
- **Dataflow Analysis:** Finding reachable states in control flow graphs
- **Abstract Interpretation:** Analyzing program properties using abstract domains

Mathematical Applications:

- Fixed points of increasing operators in Banach lattices
- Existence of solutions to functional inequalities
- Properties of measure-preserving transformations

Example (Reachability in graphs):

- Let $L = 2^V$ be the powerset of vertices
- $f(S) = S \cup \{v : \exists u \in S, (u, v) \in E\}$
- The least fixed point is the set of reachable vertices

Reflexivity and Cascading Nonexpansive Maps

Theorem (Section 5.9.1: Cascading Nonexpansive Maps)

Let X be a reflexive Banach lattice. Consider a sequence of cascading nonexpansive mappings:

$$T_1, T_2, \dots, T_n : X \rightarrow X$$

where $T_{i+1}(T_i(x)) = T_i(T_{i+1}(x))$ (commuting property).

If the mappings satisfy:

- 1 Each T_i is nonexpansive: $\|T_i x - T_i y\| \leq \|x - y\|$
- 2 X is reflexive
- 3 Appropriate boundedness conditions

Then the composed operator $T = T_n \circ T_{n-1} \circ \dots \circ T_1$ has a fixed point.

Reflexivity in Banach Spaces

Definition (Reflexive Space)

A Banach space X is **reflexive** if the natural embedding

$$J : X \rightarrow X^{**}, \quad J(x)(f) = f(x),$$

is surjective (every functional on X^* is represented by an element of X).

Properties of Reflexive Spaces:

- 1 Every bounded sequence has a weakly convergent subsequence
- 2 Weak compactness is equivalent to norm compactness
- 3 Dual space is also reflexive

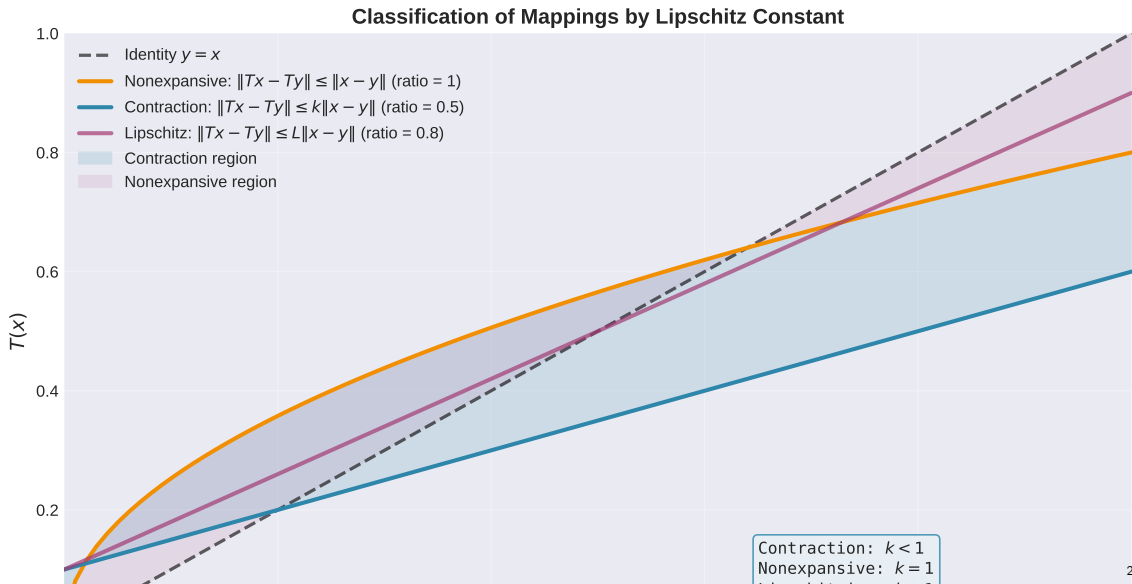
Examples:

- $\mathbb{R}^n$, Hilbert spaces ℓ^2 , $L^p(\Omega)$ for $1 < p < \infty$
- Not reflexive: c_0 , $L^\infty(\Omega)$, $C([0, 1])$

Listing: Checking reflexivity properties

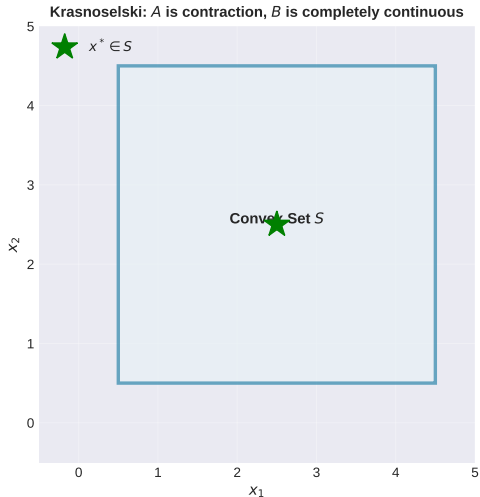
```
# Example: Banach space properties
import numpy as np
```

Lipschitz Constant and Contractiveness



Krasnoselski and Burton in Detail I

Krasnoselski and Burton in Detail II



$\mathcal{P}$ -Lipschitzian Mappings

Burton's Modification of Krasnoselski

Krasnoselski (1964):

- A : contraction on S
- B : completely continuous
- $Ax + By \in S$

Burton (1995) weakened:

- A : still contraction
- B : maps S to bounded sets
- Weaker covering condition

Both guarantee fixed points for operator equation: $Ax + Bx = x$

$\mathcal{D}$ -Lipschitzian Mappings

Hammerstein Integral Equations

Problem (Hammerstein Equation)

Solve the nonlinear integral equation:

$$x(t) = f(t) + \int_0^1 K(t, s)g(s, x(s)) ds, \quad t \in [0, 1]$$

where:

- $K(t, s)$ is the integral kernel
- g is a nonlinear function
- f is a given input

Listing: Numerical solution of Hammerstein equation

```
import numpy as np
from scipy.integrate import quad

# Discretized Hammerstein equation
def hammerstein_operator(x, kernel, g, f, t_points):
    """
    Compute  $Ax + Bg(x)$  where:
```

Summary of Key Results

Theorem	Equation	Conditions	Conclusion
Krasnoselski (5.176)	$Ax + Bx = x$	A contraction B compact	Fixed point exists
Burton (5.177)	$Ax + Bx = x$	A contraction B bounded	Weaker conditions
$\mathcal{P}$ -Lipschitz (5.184)	$Ax + Bx = x$	A $\mathcal{P}$ -Lipschitzian B compact	Fixed point exists if $M\phi(r) < r$
$\mathcal{D}$ -Lipschitz (5.185)	$AxBx = x$	A $\mathcal{D}$ -Lipschitzian B compact	Fixed point exists if $M\phi(r) < r$
Knaster-Tarski (5.9)	$f(x) = x$	f monotone complete lattice	Fixed point set is a lattice

Practical Applications

In Fixed Point Theory:

- Finding equilibrium points in dynamical systems
- Solving nonlinear equations with operator decomposition
- Existence of solutions to integral and differential equations

In Numerical Analysis:

- Fixed point iteration methods (Newton's method variants)
- Successive approximation schemes
- Convergence analysis of iterative algorithms

In Optimization:

- Proximal point algorithms
- Gradient descent and variants
- Splitting methods for composite problems

In Game Theory and Economics:

- Existence of Nash equilibria
- Fixed points in best-response dynamics
- Equilibrium in market models

Computational Implementation

Listing: Generic fixed point iterator with monitoring

```
import numpy as np
from numpy.linalg import norm

class FixedPointIterator:
    """Generic fixed point iteration method"""

    def __init__(self, operator, tol=1e-10, max_iter=1000):
        self.operator = operator
        self.tol = tol
        self.max_iter = max_iter
        self.history = []

    def iterate(self, x0):
        """Run fixed point iteration"""
        x = x0.copy()
        for n in range(self.max_iter):
            x_old = x.copy()
            x = self.operator(x)
            error = norm(x - x_old)
            self.history.append(error)

            if error < self.tol:
                print(f"Converged in {n} iterations")
                return x, self.history
```

Convergence Verification

Listing: Verifying theoretical convergence rates

```
import numpy as np
import matplotlib.pyplot as plt

# Simulate convergence for different methods
def simulate_convergence(contraction_ratio, n_iter=30):
    """Simulate fixed point iteration convergence"""
    errors = np.zeros(n_iter)
    errors[0] = 1.0 # Initial error
    for n in range(1, n_iter):
        errors[n] = contraction_ratio * errors[n-1]
    return errors

# Compare different contraction ratios
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4))

# Linear scale
for k in [0.3, 0.5, 0.7, 0.9]:
    errors = simulate_convergence(k)
    ax1.plot(errors, 'o-', label=f'$k={k}$', markersize=4)
ax1.set_xlabel('Iteration $\mathbf{n}$')
ax1.set_ylabel('Error $\mathbf{e_n}$')
ax1.set_title('Linear $\mathbf{Scale}$')
ax1.legend()
ax1.grid(True, alpha=0.3)
```

Section 5.8: Fixed Point Theorems in Banach Algebra

- ① **Krasnoselski (1964):** Decompose $T = A + B$ where A is contraction, B is compact
- ② **Burton (1995):** Weaker hypothesis replacing complete continuity
- ③ **$\mathcal{P}$ -Lipschitzian (5.184):** Generalized contractions with function ϕ
- ④ **$\mathcal{D}$ -Lipschitzian (5.185):** Further generalization (Dhage 2003)

Section 5.9: Lattice-Theoretic Fixed Point Theorems

- ① **Knaster-Tarski (1927):** Monotone functions on complete lattices have fixed points
- ② **Fixed Point Lattice:** Set of fixed points forms a complete lattice
- ③ **Cascading Nonexpansive Maps:** Compositions preserve fixed points in reflexive spaces
- ④ **Reflexivity:** Key property ensuring weak compactness in infinite dimensions

Key Theorems Summary III

Unifying Concepts:

- **Contractiveness:** Core property enabling guaranteed convergence
- **Compactness:** Enables treating perturbations
- **Monotonicity:** Sufficient for lattice fixed points without continuity
- **Decomposition:** Breaking complex problems into simpler components

Extensions and Further Reading

Advanced Topics:

- Multivalued mappings and fixed points
- Fixed points in metric spaces and ultrametric spaces
- Approximate fixed points and ε -fixed points
- Fixed point theorems in partial metric spaces

Computational Aspects:

- Iterative schemes: Mann, Ishikawa, Krasnoselski iterations
- Convergence rates and acceleration techniques
- Parallel and distributed fixed point algorithms

Applications:

- Differential equations: Picard-Lindelöf theorem, periodic solutions
- Integral equations: Hammerstein, Fredholm, Volterra equations
- Variational inequalities and complementarity problems
- Machine learning: Neural network fixed points, equilibrium models

Conclusion

Main Takeaways:

- ① **Decomposition:** Complex problems can be split into manageable parts
 - Contraction + compact perturbation (Krasnoselski-Burton)
 - Monotone increasing functions on lattices (Knaster-Tarski)
- ② **Generalization:** From strict contractions to Lipschitzian and $\mathcal{D}$ -Lipschitzian mappings
- ③ **Abstraction:** Lattice structures provide algebraic framework without topology
- ④ **Computability:** Iterations provide constructive proofs and practical algorithms

Next Steps:

- Apply to specific problem domains (differential equations, optimization)
- Study convergence rates and complexity
- Explore parallel and distributed implementations